

The prevention project's health measure

What it is made of, how it is built, and what it delivers

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Built by `data_cleaning/04.build_health.py` over `measures/health.py`; the figures and the properties table by `descriptives/21.health_measure.py`. Every number below is printed by those two scripts. The search that chose this measure among 85 candidates is archived: `health_measures_note.tex` and `redo_concepts/baseline_measure_comparison.xlsx`.

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1 The measure in one page

The project needs one number per person-wave saying how healthy someone is *physically*, on a scale where being sicker costs more at an increasing rate, where the very unwell are not all bunched at a floor, and where the answer does not turn on how the scale happens to be cardinalised. That number is fitted to eight ordinal items from UKHLS:

Item	What it records	Categories
GH	self-rated general health (sf1)	5
PF	moderate activities and climbing stairs (sf2a , sf2b)	5
RP	accomplished less, limited in kind of work (sf3a , sf3b)	9
BP	pain interference with normal work (sf5)	5
LIM_FL	functional limitations: mobility, lifting, dexterity, co-ordination	5
LIM_SELF	self-care: personal care, continence	3
LIM_SENS	sensory: hearing, sight	3
COND	ever-diagnosed physical conditions, 0–5 and 6 or more	7

The first four are the SF-12 physical testlets already validated for this project; the next three count the UKHLS impairment areas, grouped; the last counts diagnoses. In the search’s notation this bank is **P-LIM3+CC**. The sample is **387,599 person-waves** on **69,958 people**, waves 1–15, ages 20–90, complete on all eight items — 5.5 observations per person on average.

A graded-response model turns the eight answers into a latent health score. The measure is reported three ways, all monotone in the same answers:

	Scale	What it is
θ	standard normal in the pool	the latent score, expected a posteriori
h	$[0, 1]$, here 0.11 to 0.98	θ through the expected-score curve: the share of the bank’s total scale a person is expected to answer at
fi10	$[0, 1]$, 1 = no deficits	a ten-deficit index: the six SF-12 physical items graded by response step, the three impairment groups as the share of their areas, and the condition count capped at six — equal weights, computable by hand

They rank people almost identically (Spearman 0.991 between h and **fi10**) and they agree on every conclusion the paper draws, with one exception that matters and is set out in section 5.3: how much of the cost gradient’s apparent age-invariance is a property of the scale rather than of the data. θ is the default; h is the scale the framework’s $[0, 1]$ health variable maps to; **fi10** is the transparent version a reader can reconstruct from the questionnaire.

2 The variables and the coding rules

2.1 The SF-12 physical testlets

Six SF-12 items enter as four testlets, the grouping validated in the archived note: the two physical-function items sum into PF, the two role-physical items sum into RP, and general health and pain enter alone. Summing the pairs is what removes the positive local dependence between them — the two items in each pair share wording, and left separate their residual correlation would be read as evidence of a second dimension. Every item is coded so category 1 is worst health.

2.2 The impairment areas

UKHLS asks which of a list of areas a person’s health limits them in. Eight are physical and used here, in three groups:

Group	Areas (<code>disdif</code> codes)	Count
LIM_FL	mobility (1), lifting (2), dexterity (3), co-ordination (10)	0–4
LIM_SELF	continence (4), personal care (11)	0–2
LIM_SENS	hearing (5), sight (6)	0–2

Two coding rules hold throughout, and both are decisions rather than data.

The routing rule. In the early waves the battery was asked only of people reporting a long-standing illness. Someone who reports none therefore has no answer, not a zero. We count them as having no limitation, in every wave, so that the variable means the same thing across the panel; someone who does report a long-standing illness but has no usable answer is missing. The alternative — treating unrouted rows as missing — would drop the healthy half of the sample, and with it the part of the distribution where prevention happens.

The merge rule. A top count category holding less than 0.5% of person-waves is merged into the one below, decided on the sample before any model is fitted. This is what fixes LIM_FL at five categories (0, 1, 2, 3, 4+) and LIM_SELF and LIM_SENS at three. Categories thinner than that are estimated off a few hundred people and their thresholds wander.

Two areas are deliberately left out. “Recognising when in physical danger” loads with the mental items, and the whole point of this measure is that it carries no mental health. “Other health problem or disability” is non-specific: added as its own group (the search’s P-LIM3+O) it changes nothing, and it is one more judgement call for no gain.

2.3 The condition count

COND counts the sixteen ever-diagnosed physical conditions in the UKHLS inventory, 0 to 5 and 6 or more, the top category set by the merge rule on the sample that has an inventory. It enters as one item, not as six clinical-group items. Both versions were fitted. As six items the conditions take 40% of the weight in an unweighted index against 14.1% in the model, so the model version and the by-hand version stop agreeing — which defeats the purpose of having a by-hand version at all. As one count they take 6% in the model and 10% in `fi10`, and the three variants stay interchangeable.

Requiring an inventory is what sets the sample: 387,599 person-waves rather than the 497,423 that have SF-12 and limitations. That restriction is shared with every condition-carrying candidate in the search, so it changes no comparison, and it leaves the cost and mortality samples intact.

2.4 Category use

	k	share of person-waves, best category first (%)								
GH	5	13.29	33.33	31.71	15.58	6.10				
PF	5	65.13	11.66	11.82	4.60	6.79				
RP	9	52.18	6.85	12.77	4.55	11.57	2.68	4.76	1.20	3.44
BP	5	53.72	25.24	9.14	8.26	3.64				
LIM_FL	5	83.05	6.66	5.30	3.24	1.76				
LIM_SELF	3	94.15	4.74	1.12						
LIM_SENS	3	95.62	3.82	0.56						
COND	7	48.28	28.88	12.90	5.59	2.44	1.05	0.86		

7.1% of person-waves give the best answer to all eight items. That is the measure’s ceiling, and it is not a problem for this project: prevention questions are asked at the other end, where 12 person-waves sit at the worst answer on everything.

3 The model

3.1 Graded response

Item j with k_j categories has one discrimination a_j and $k_j - 1$ thresholds $b_{j1} < \dots < b_{j,k_j-1}$. The probability of answering at or above category c is logistic in θ ,

$$P(Y_j \geq c \mid \theta) = \frac{1}{1 + \exp\{-D a_j(\theta - b_{j,c-1})\}}, \quad D = 1.702,$$

and a single category’s probability is the difference of adjacent curves. The model is Samejima’s; it is the ordinal factor model in another parameterisation (Takane and de Leeuw, 1987), which is where the weighted sum in appendix A comes from.

Estimation is marginal maximum likelihood by EM over distinct answer patterns (28,360 of them), with the latent distribution identified as $N(0, 1)$ in the pooled sample and 121-point Gauss quadrature on $[-6, 6]$. Fitting took 60 EM iterations. Scores are expected a posteriori under the pooled prior, so a θ means the same thing at every age — the age profile is estimated separately and does not feed back into anyone’s score.

Item	k	a	thresholds							
GH	5	2.03	-2.10	-0.98	0.16	1.45				
PF	5	3.52	-1.73	-1.38	-0.80	-0.40				
RP	9	3.09	-2.20	-2.03	-1.56	-1.37	-0.81	-0.64	-0.23	-0.03
BP	5	2.31	-2.35	-1.50	-0.99	-0.09				
LIM_FL	5	3.38	-2.49	-1.94	-1.46	-1.07				
LIM_SELF	3	2.43	-2.95	-1.99						
LIM_SENS	3	1.36	-4.49	-2.85						
COND	7	1.18	-4.61	-3.90	-3.12	-2.26	-1.25	0.11		

The pattern is the one the measure was built for. PF, RP and LIM_FL discriminate hardest and their thresholds sit between -2.5 and 0 : they are what separates people in the unhealthy half. GH and BP discriminate less but reach further up, so they keep the healthy half from collapsing onto one point. LIM_SENS and COND discriminate least and sit furthest out — they are the items that keep the very sick from piling up at a floor, which is exactly the job the archived search hired them for.

3.2 Why the graded-response model and not the partial-credit one

Both were fitted, on all six limitation banks. The graded-response model wins by 13,500 to 15,200 log-likelihood points at equal parameter counts, every time. The reason is structural: the generalised partial credit model makes a weighted sum of category scores a sufficient statistic, which forces its steps to be ordered in the data — and PF and RP are sums of two items whose categories are used very unevenly, so their steps are not. Their θ s correlate at 0.995 and their clusters agree 97% of the time, so nothing here turns on the choice; the graded-response model is simply the better-fitting one.

3.3 Local dependence

Yen's Q_3 on the eight testlets has no positive off-diagonal above $+0.052$ (the expected value under the model is about $-1/(J-1) = -0.14$). Nothing in the bank is measuring a second thing.

3.4 The age profile

Refitting multigroup by single year of age — item parameters common across ages, the pooled latent distribution standardised — gives the latent mean and standard deviation at each age. Mean health falls by 1.23 pooled standard deviations between 30 and 80, and the spread widens with age (figure 1c). These are descriptive: no score uses them.

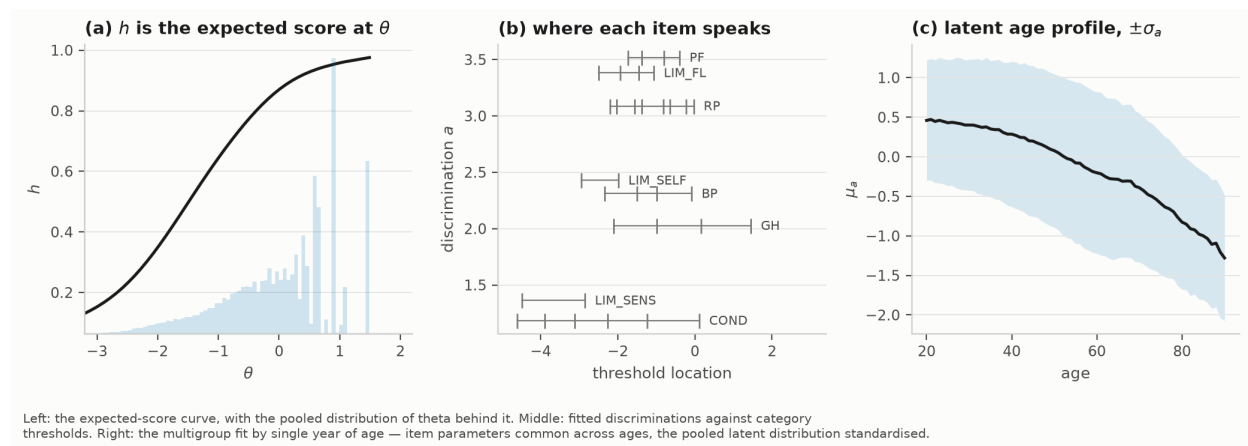


Figure 1: The measure. (a) the expected-score curve that turns θ into h , over the pooled distribution of θ ; (b) each item's discrimination against the location of its thresholds; (c) the latent age profile.

4 The three variants

4.1 θ and h

h is θ pushed through the test characteristic curve and rescaled to $[0, 1]$: the expected share of the bank's total scale. The map is monotone, so θ and h order people identically and no rank statistic can separate them. It is not affine, and that is the whole of the difference between them: the curve is flat at the top and steep in the middle (figure 1a), so a standard deviation of θ buys much less h among the healthy than among the sick. Anything reported per standard deviation — a cost gradient, a variance profile — therefore depends on which of the two is used. Section 5.3 is about exactly that.

4.2 The ten-deficit index

`fi10` is the version that needs no model. Each of ten deficits is scored between 0 and 1 and the index is one minus their mean, so 1 is no deficits:

6 deficits	the SF-12 physical items <code>sf1</code> , <code>sf2a</code> , <code>sf2b</code> , <code>sf3a</code> , <code>sf3b</code> , <code>sf5</code> , each graded 0 to 1 by response step
3 deficits	<code>LIM_FL</code> , <code>LIM_SELF</code> , <code>LIM_SENS</code> as the share of their areas mentioned
1 deficit	the condition count, capped at six, divided by six

This follows the deficit-accumulation tradition (Mitnitski, Mogilner and Rockwood, 2001; Searle et al., 2008), with one departure that is not cosmetic: the impairment *areas* are grouped before they are averaged. Scored area by area, the eight areas would take more than half the index's weight and the result stops being convex in cost — on costs capped at the 99th percentile the area version's cost curvature is $t = -2.0$ against $t = +12.0$ for the grouped one. Equal weights are defensible here only because the groups make the units comparable first.

`fi10` takes 121 distinct values, against 28,360 for θ . It is coarser, not discrete in any way that matters: no value holds a large mass and the bottom percentile alone is spread over 26 of them.

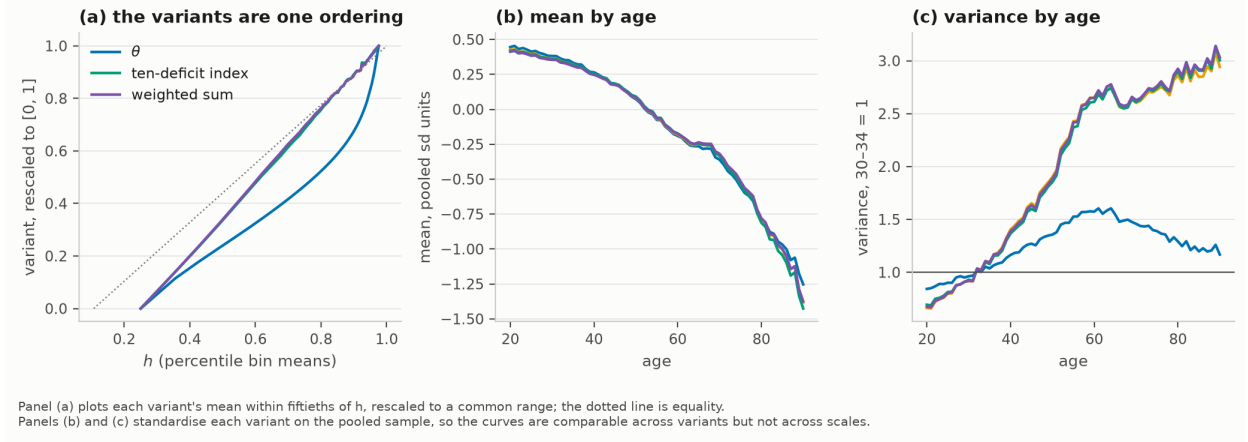


Figure 2: The variants. (a) each variant's mean within fiftieths of h , rescaled to a common range — fi10 and the weighted sum track h almost exactly, while θ bends away because the map between them is not affine; (b) and (c) mean and variance by age, each variant standardised on the pooled sample.

5 What the measure delivers

The criteria below are the ones the search was run against; the archived note has them for all 85 candidates.

5.1 No floor, and no discreteness that bites

	distinct	at the worst	variance ratio		cost curvature	
	values	value, 60–84	30→60	60→85	$\hat{\gamma}$	t
θ	28,360	7	1.62	0.77	105.5	36.9
h	28,360	7	2.85	1.08	34.6	11.3
fi10	121	7	2.80	1.12	36.8	11.9
weighted sum	28,360	7	2.84	1.11	30.2	9.9

Seven person-waves in the whole 60–84 sample sit at the worst attainable score — 0.006%. The same share is 1.11% for the functioning-only bank the project used before, 0.10% once the sensory and continence areas are added and 0.023% for the three-group version without conditions. The condition count is what finishes the job: it keeps separating people after every limitation item has bottomed out.

5.2 Convexity in cost

Cost rises with poor health at an increasing rate on every variant: the quadratic term in the standardised measure is positive with t between 9.9 and 36.9, person-clustered, on costs capped at the 99th percentile (figure 3). This is the criterion the whole search was really about, and it is the one that fails for plausible-looking alternatives: an ungrouped deficit index is *concave*.

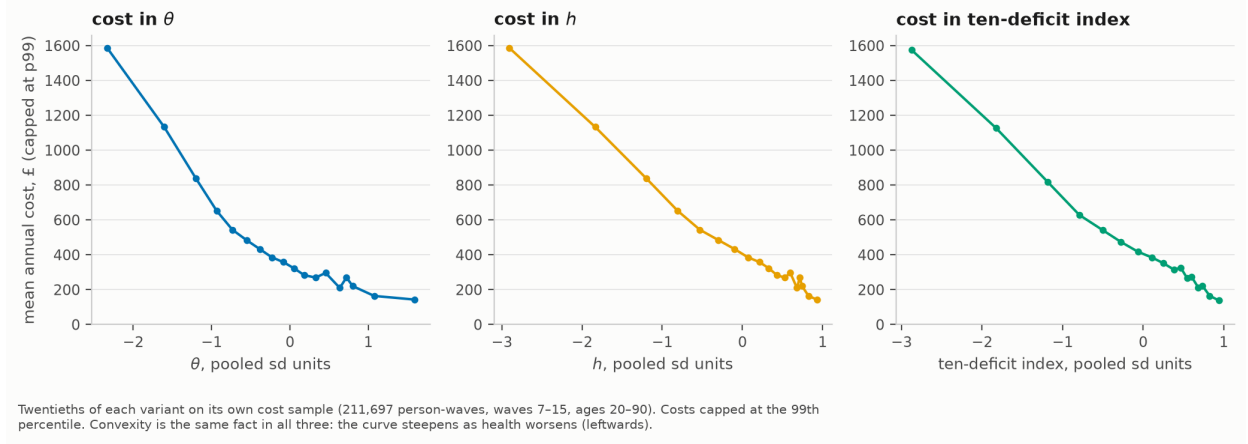


Figure 3: Mean annual healthcare cost by twentieth of each variant. The curve steepens as health worsens (leftwards) in all three.

5.3 Age-invariance is a property of the scale, not of the data

The proportional cost gradient — log points of cost per standard deviation sicker, within an age band — looks very different on the two model scales:

	20–34	75–90	ratio	$z \times \text{age}, t$
θ	0.918	0.883	0.96	3.5
h	1.124	0.729	0.65	10.2
fi10	1.059	0.726	0.69	6.9

On θ the gradient is nearly the same at 80 as at 25; on h and on the deficit index it is a third smaller. Both are correct. A standard deviation is not the same amount of health on the two scales, and the expected-score curve stretches the unhealthy end, where the old are. The cost proxy plan’s statement that “the health–cost relationship is close to age-invariant” was computed on θ and reproduces exactly; it should be read as a statement about θ and not as a scale-free fact.

The scale-free version of the same claim survives: fix a health decile and cost barely moves with age. In the sickest decile of the pooled health distribution, mean cost is £2,143 at 20–34 against £2,136 at 75–90, while mean cost over everyone rises 2.6-fold across the same span (£374 to £969). The substantive reading — that what the returns to prevention depend on is health, not age, conditional on health — holds. What does not hold is any claim that “per standard deviation” means one thing.

5.4 Identification

The framework’s identification tests ask whether the covariance structure of four ages two years apart can separate true health dynamics from measurement error, and in particular whether $\text{Corr}(H, d)$ follows the sign path the model implies — positive among the young, negative among the old. The tests differ in what they assume the error looks like and in which moments they use:

	balanced moments			pairwise moments		
	Case 2	Case 3	Case 4	Case 2	Case 3	Case 4
θ	✓	✓	✓	✓	—	✓
h	✓	✓	✓	✓	—	—
fi10	—	✓	✓	✓	—	—
weighted sum	✓	✓	✓	✓	—	—

Case 2 is AR(1) measurement error, Case 3 adds age-specific error variance, Case 4 is AR(1) plus a one-period spike (evaluated at the best admissible persistence). θ recovers the sign path under five of the six; every variant recovers it under at least three, including Case 4, which is the most permissive about what the noise can be. Case 3 on pairwise moments fails for all 85 candidates in the search, this measure included — it is a property of that estimator, not of any measure.

5.5 Mortality

The log-odds of dying before the next wave fall by 1.13 per standard deviation of θ (0.89 on h , 0.90 on **fi10**); the death rate in the least healthy decile is 43 times that in the healthiest.

5.6 A simpler version reaches the same conclusions

This was a requirement, not a bonus. The same eight variables, pooled three ways — the model’s θ , its $[0, 1]$ transform, and an equal-weighted deficit index — agree on the floor, on continuity, on convexity, on the variance profile, on mortality and on the sign path, with the single scale-dependent exception in section 5.3, which is a fact about cardinalisation that we now state rather than a disagreement about the world. The appendix weighted sum makes the same point inside the model: h is a weighted sum of the eight coded answers to $R^2 = 0.993$.

6 Reproducing it

<code>data_cleaning/01.build_panel.py</code>	the processed UKHLS panel
<code>02.build_measures.py</code>	the SF-12 item extract every later step reads
<code>03.build_chronic.py</code>	condition histories
<code>04.build_health.py</code>	the measure: <code>health_measure.parquet</code> with <code>theta</code> , <code>theta_sd</code> , <code>h</code> , <code>fi10</code> , <code>ws</code> , plus item, Q_3 , age-profile and weight tables
<code>05.build_cost_proxy.py</code>	the cost index
<code>descriptives/21.health_measure.py</code>	this note’s figures and properties table

The bank, the coding rules and the deficit index live in `measures/health.py`, so anything that needs the measure’s definition imports it rather than re-deriving it. `04.build_health.py` gates every run on: no positive local dependence, the EAP identity $\text{Var}(\theta) + \overline{\text{SE}^2} = 1$ (it holds to three decimals), the three variants agreeing in rank, and exact reproduction of the search’s P-LIM3+CC scores while those outputs are still on disk. Outputs derive from licensed UKHLS microdata and are gitignored.

A The weighted sum behind h

Regressing h on the eight standardised category codes gives $R^2 = 0.993$. Normalised to sum to one, the weights are:

	GH	PF	RP	BP	LIM_FL	LIM_SELF	LIM_SENS	COND
weight (%)	13.1	22.1	22.0	13.5	15.2	5.4	2.5	6.3

This is not a coincidence of this sample. In the ordinal factor model that the graded-response model is a reparameterisation of, the loading is $\lambda_j = a_j / \sqrt{a_j^2 + 1.702^2}$ and the optimal score weights are proportional to λ_j / ψ_j — so a graded-response score is, to a very good approximation, a weighted sum whose weights the data chose. The practical content of the model is therefore the *weights*, not the functional form: it puts 71% on the SF-12 testlets, 23% on the impairment groups and 6% on diagnoses, where fi10 puts 60%, 30% and 10%. That the two give the same answers (Spearman 0.991) is what makes the model defensible to an audience that does not want one.

B Relation to what is used elsewhere

The SF-12 physical component summary is itself a weighted sum of these items, with weights from a 1990s US norming sample and a factor structure that mixes in the mental items; the archived note documents a 6.6-point artifact at the median physical profile that follows from that mixing. The graded-response approach differs in three ways: the weights are estimated on this sample, the mental items are absent by construction, and the scale is a latent one rather than a linear combination with an arbitrary origin.

Item-response measurement of physical function is standard in the health measurement literature (PROMIS physical function, the SF-36 IRT literature) and deficit accumulation is standard in the frailty literature; the combination of SF-12 items with impairment counts and a diagnosis count is ours, and is documented here rather than borrowed. In economics the closest relatives are the health indices built by factor methods from panel surveys — the point of appendix A is that a reader who prefers that tradition can read our measure as one of those, with weights in a table, and lose nothing.